

L1: Introduction

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search ID: gca0325

"We call him the semi-conductor - he only works part time."

Semiconductor Industry

Chemical Engineering

Chemistry

Material Science

Electronics Engineering

Applied Physics

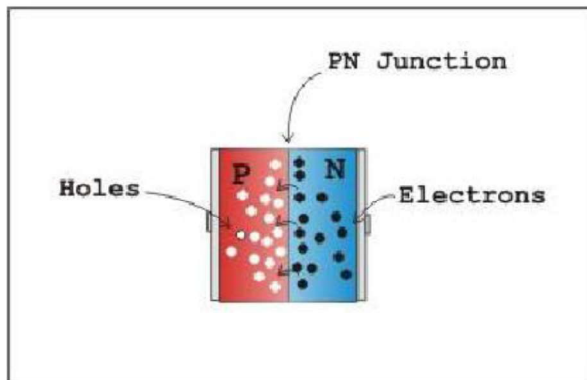
Mechanical Engineering



Truly multi disciplinary field

Devices in Action

Diode: Applications ?



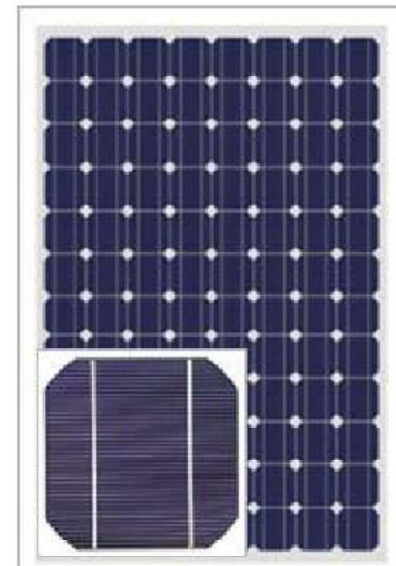
OLED Display



LED

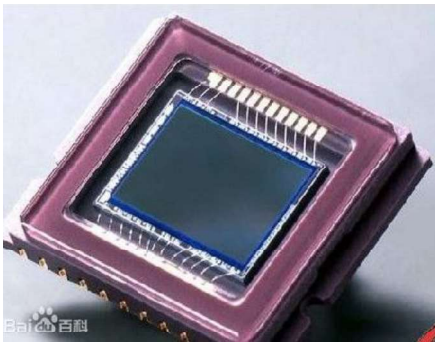
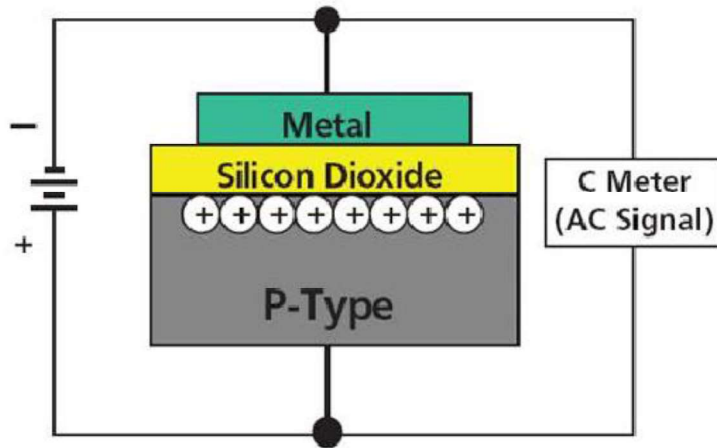


Laser Diode



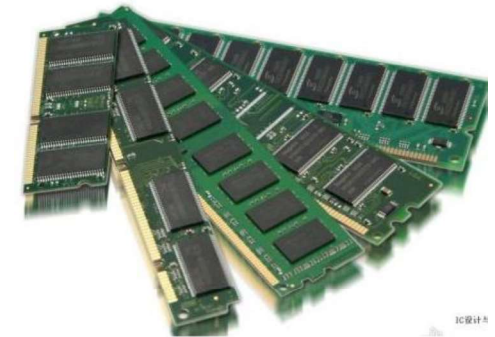
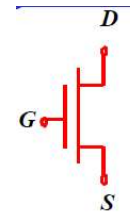
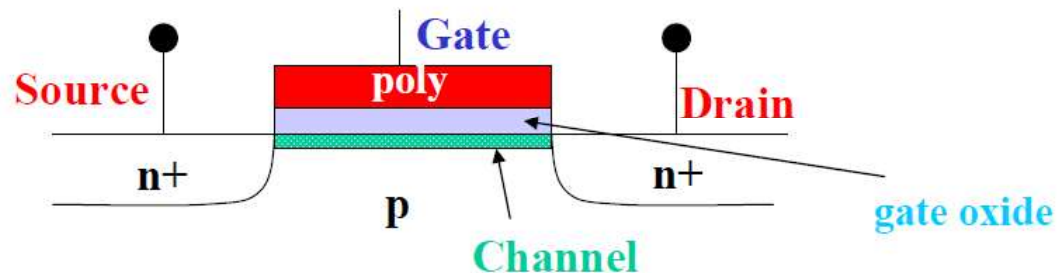
Solar Cell

MOSCAP: Applications ?

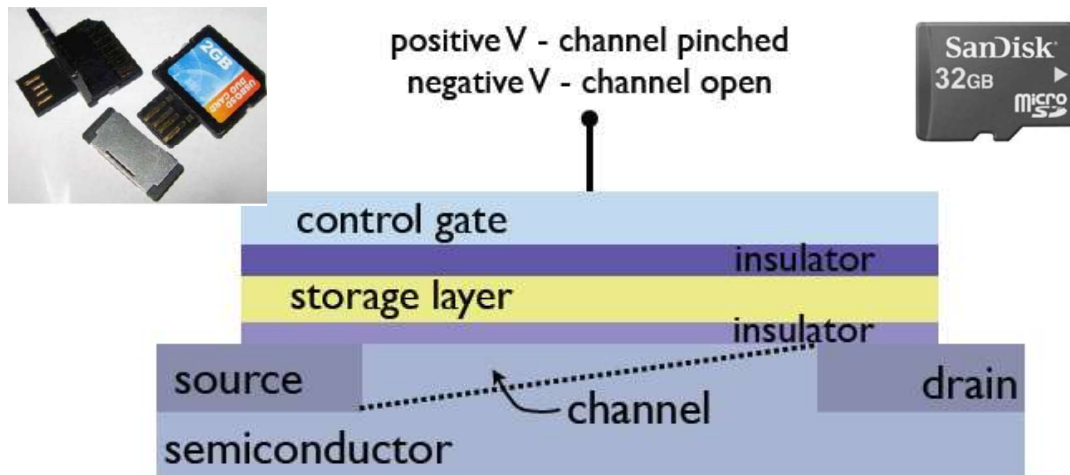


Charge Coupled Devices

MOSFET: Applications ?

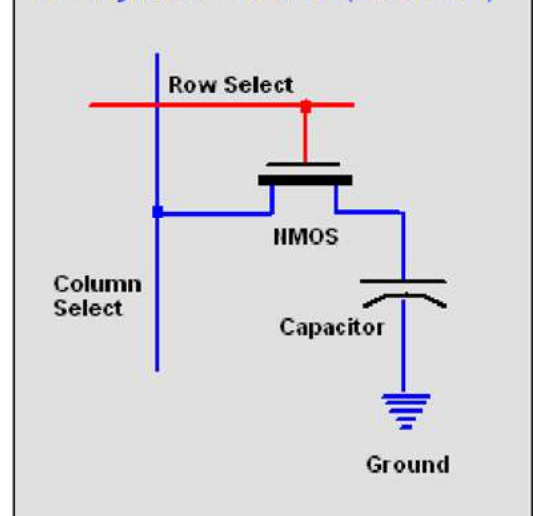


the basics of Flash



- electrons go between source and drain
- 'control gate' regulates how many negative voltage stops electrons, positive helps

One Dynamic RAM Bit (DRAM bit)

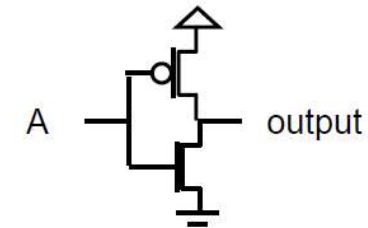


CMOS applications?

Inverter



x	F
0	1
1	0

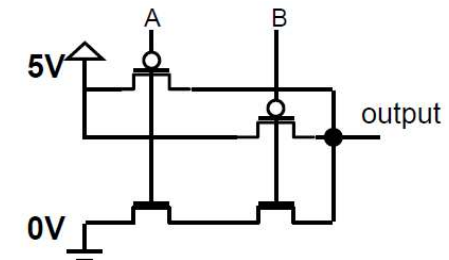


NOT Gate (Inverter)

NAND



x	y	F
0	0	1
0	1	1
1	0	1
1	1	0

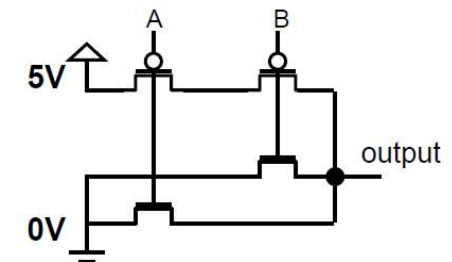


NAND Gate

NOR



x	y	F
0	0	1
0	1	0
1	0	0
1	1	0



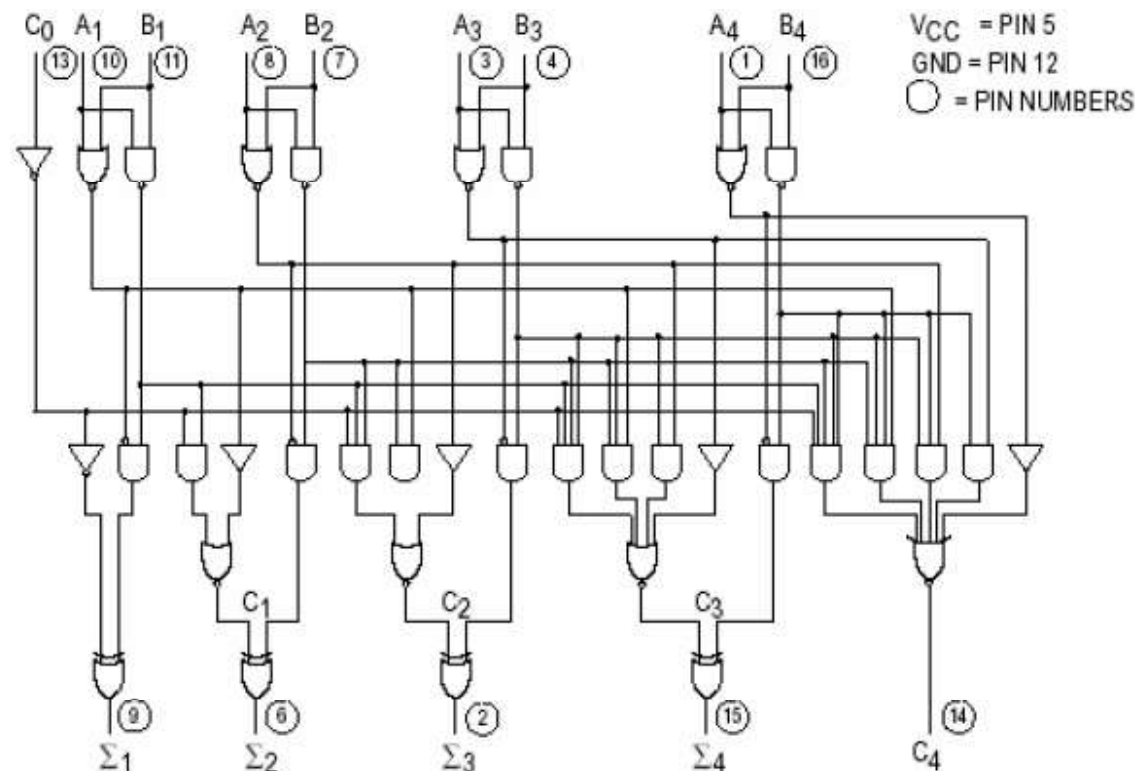
NOR Gate



Example: 4-bit Binary Adder

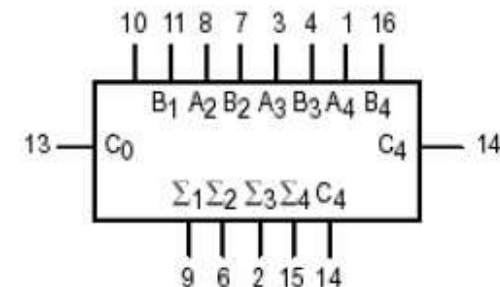
4-BIT BINARY FULL ADDER WITH FAST CARRY

The SN54/74LS83A is a high-speed 4-Bit binary Full Adder with internal carry lookahead. It accepts two 4-bit binary words (A_1 – A_4 , B_1 – B_4) and a Carry Input (C_0). It generates the binary Sum outputs Σ_1 – Σ_4 and the Carry Output (C_4) from the most significant bit. The LS83A operates with either active HIGH or active LOW operands (positive or negative logic). The SN54/74LS283 is recommended for new designs since it is identical in function with this device and features standard corner power pins.



**4-BIT BINARY FULL ADDER
WITH FAST CARRY**
LOW POWER SCHOTTKY

LOGIC SYMBOL



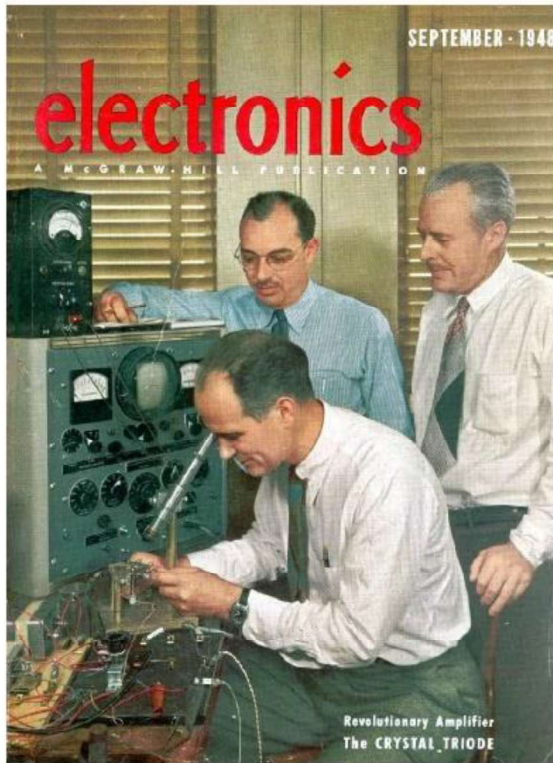
Moore's law

How old is this field ?

1st Transistor

Christmas Eve 1947

By Bardeen, Brattain, and Shockley,
Nobel Laureates in Physics 1956

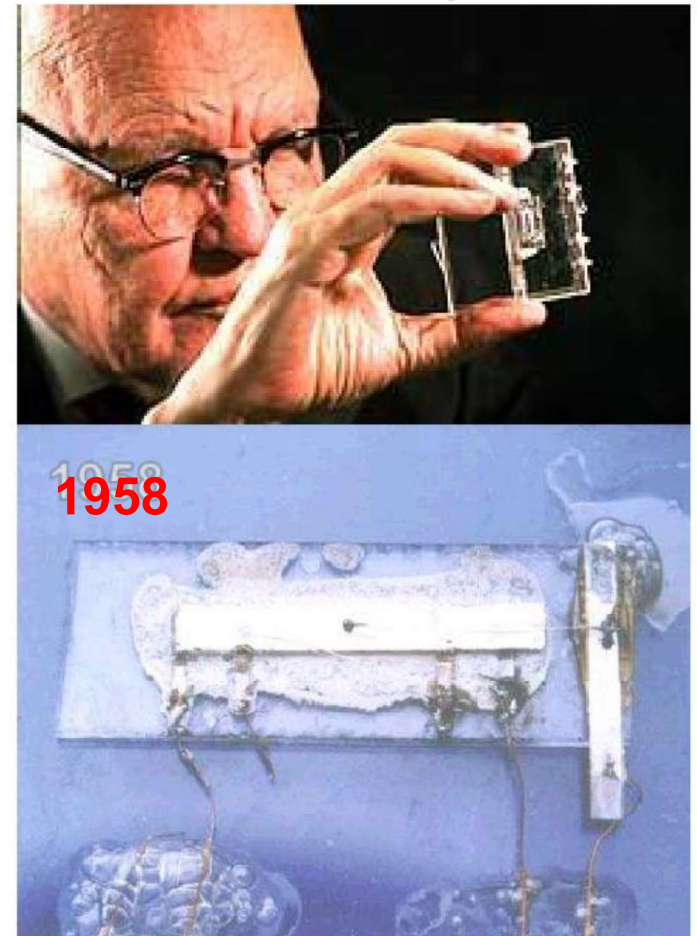


(http://www.bellsystemmemorial.com/belllabs_transistor.html)



1st Integrated Circuit

By Kilby,
Nobel Laureate in Physics 2000
and
Robert Noyce

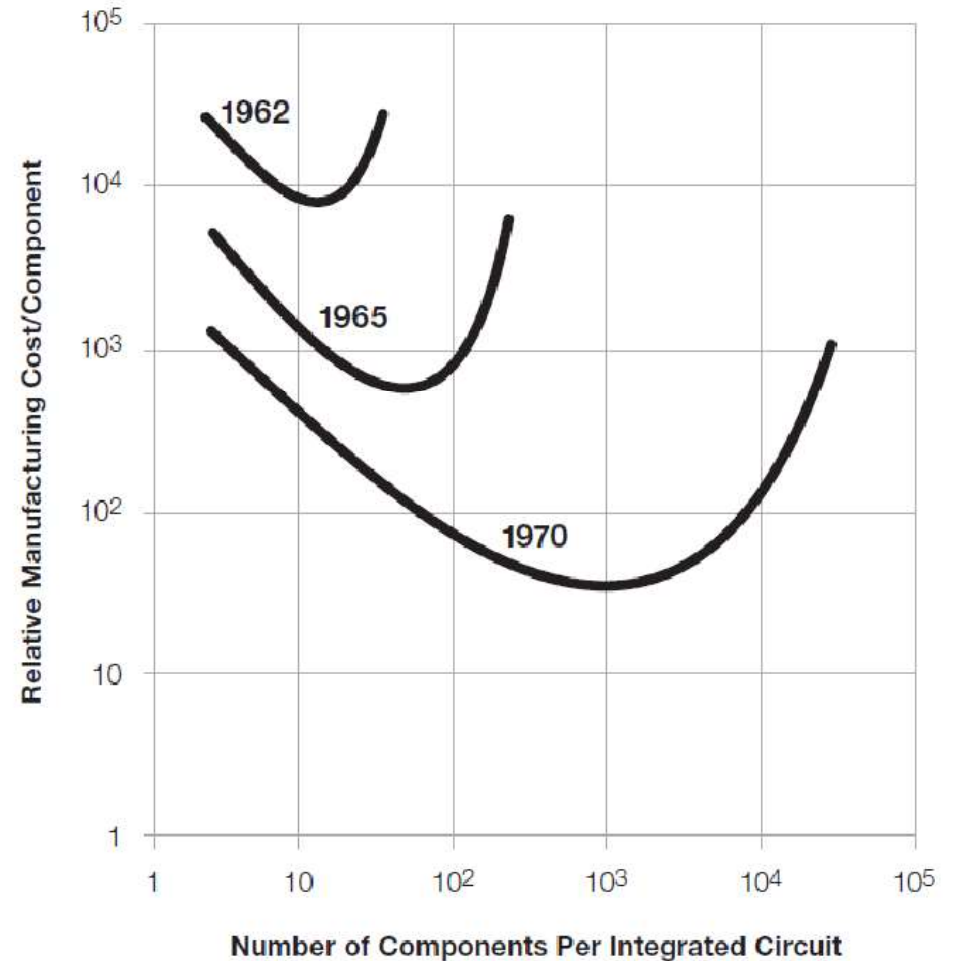
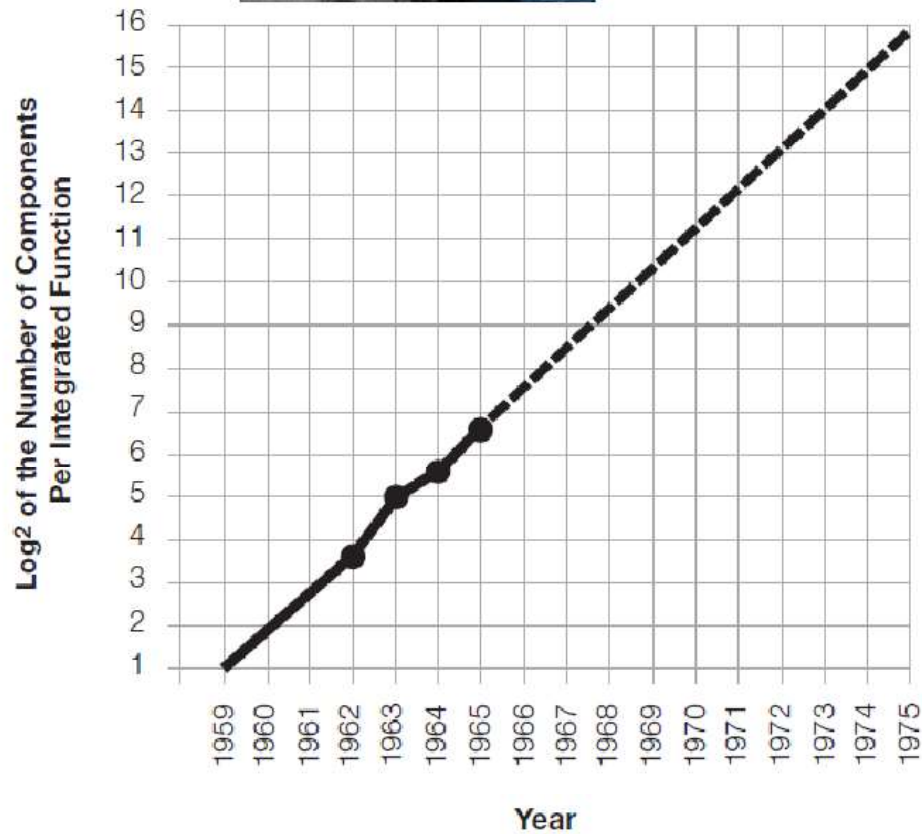


(Courtesy of TI and Huff, SEMATECH)

Gordon Moore: Electronics Magazine, 1965



The # of transistors doubles every 18 months in IC

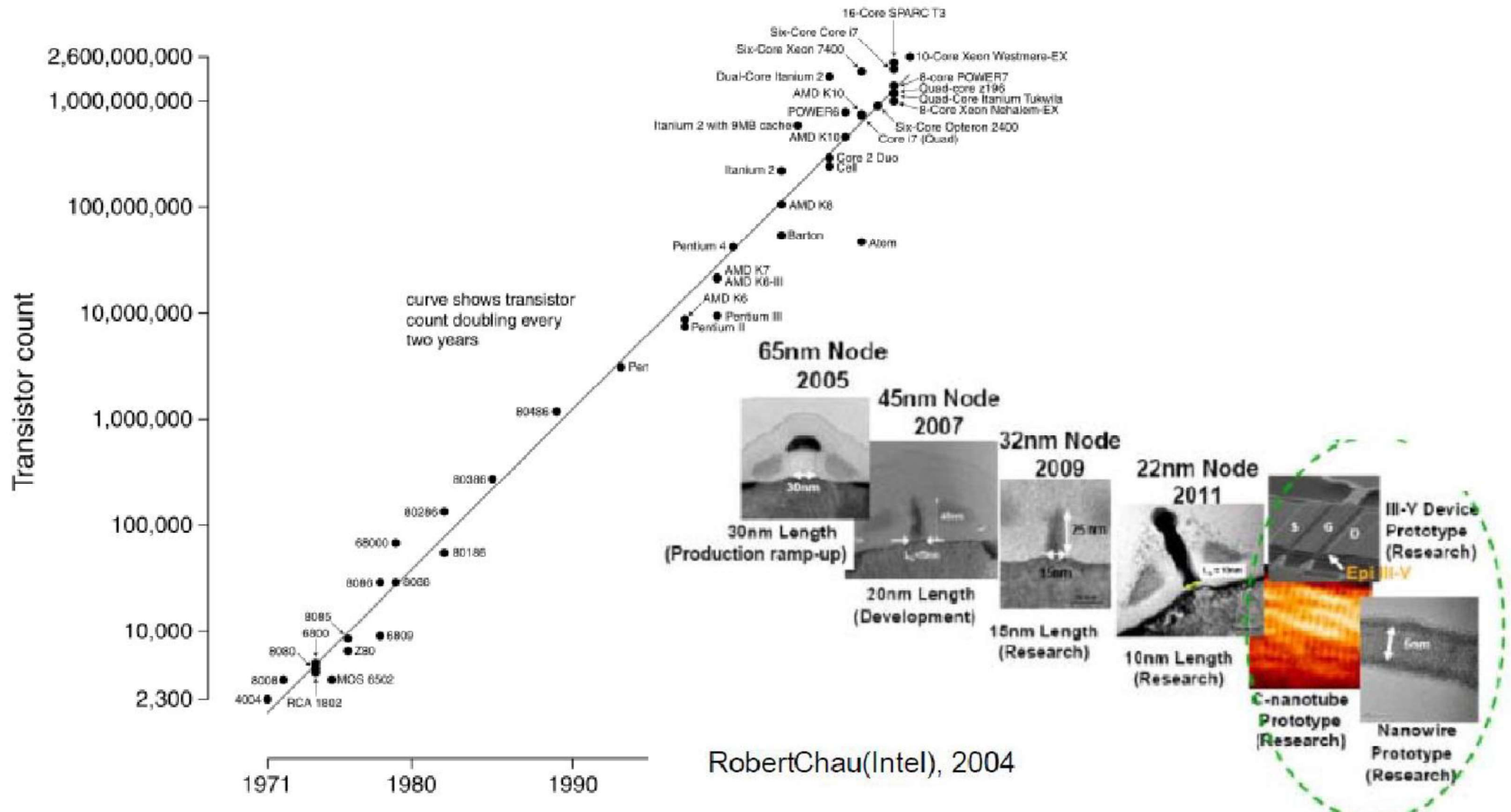


http://download.intel.com/museum/Moores_Law/Articles-Press_releases/Gordon_Moore_1965_Article.pdf

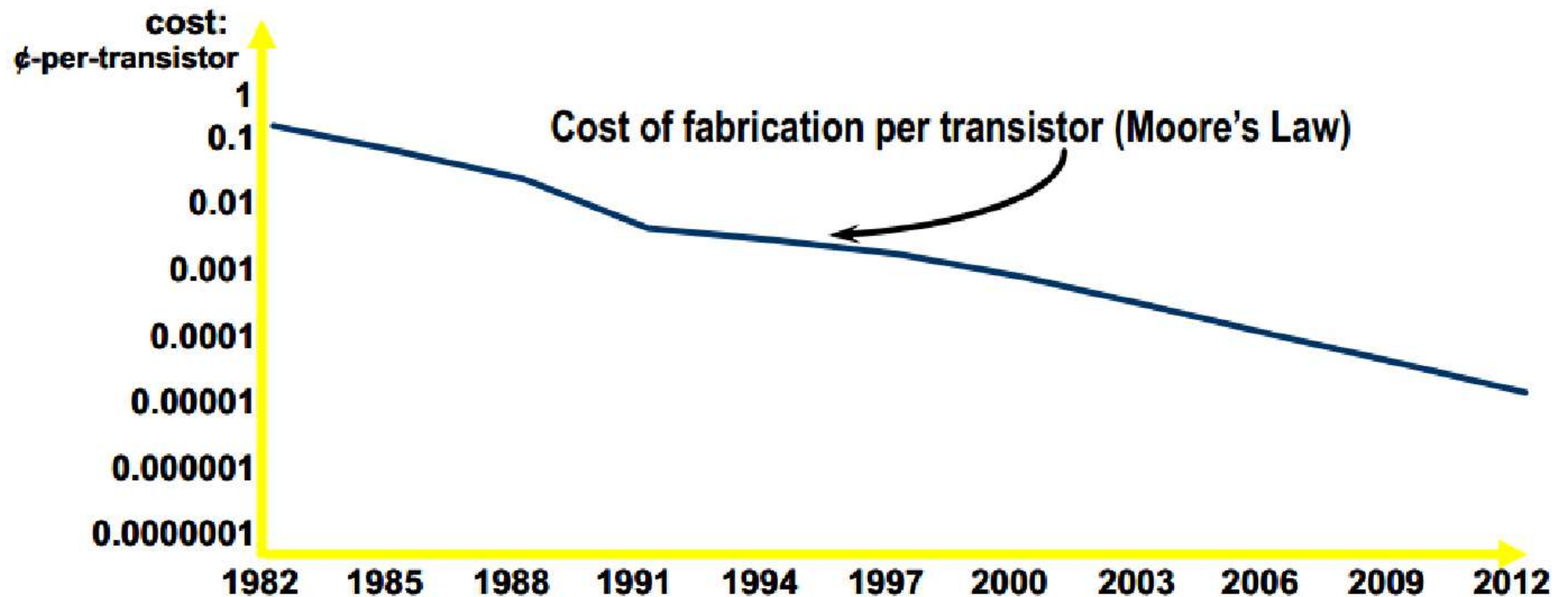
Moore's Law (what's working)

The # of transistors doubles every 18 months

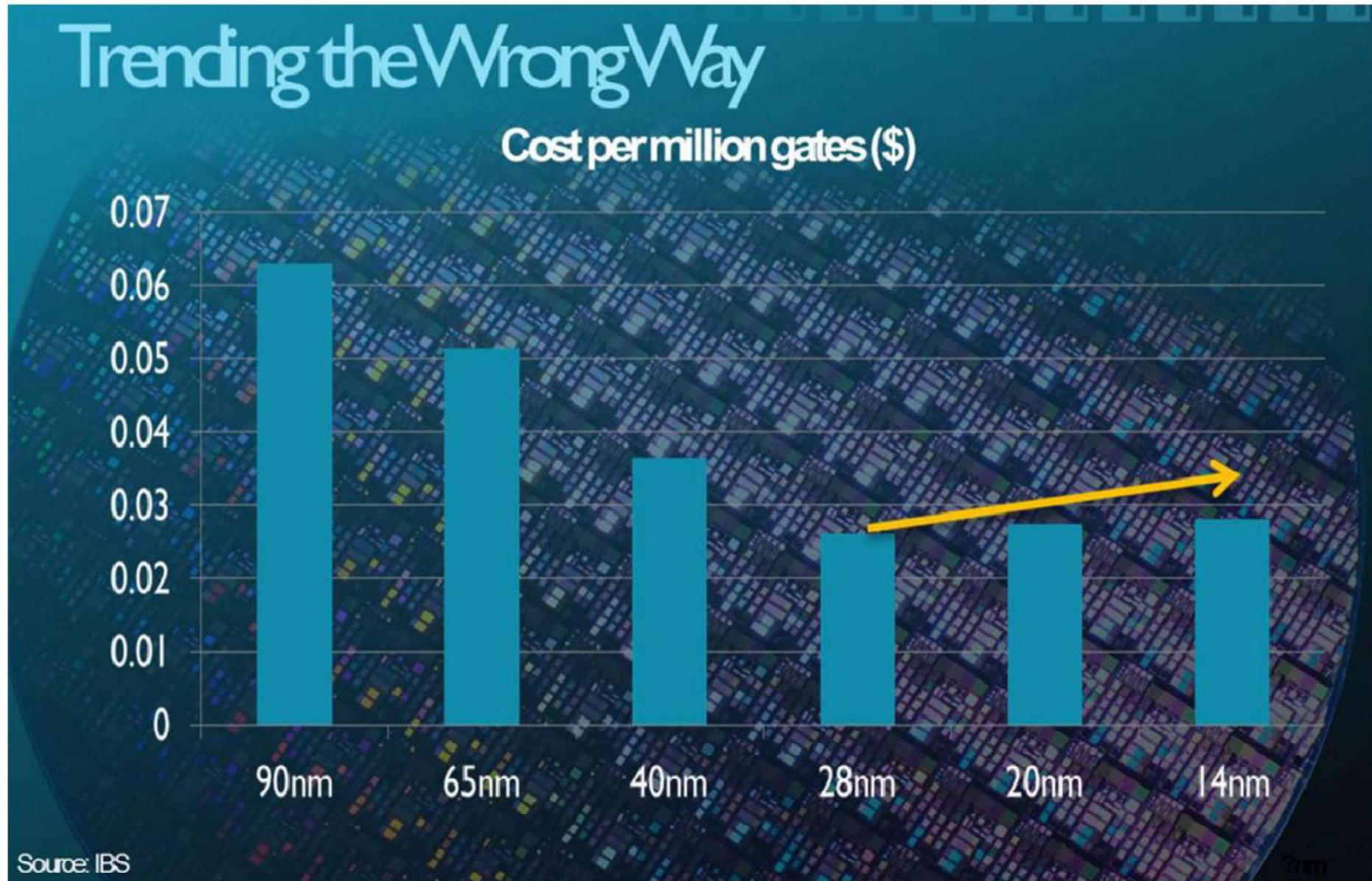
Microprocessor Transistor Counts 1971-2011 & Moore's Law



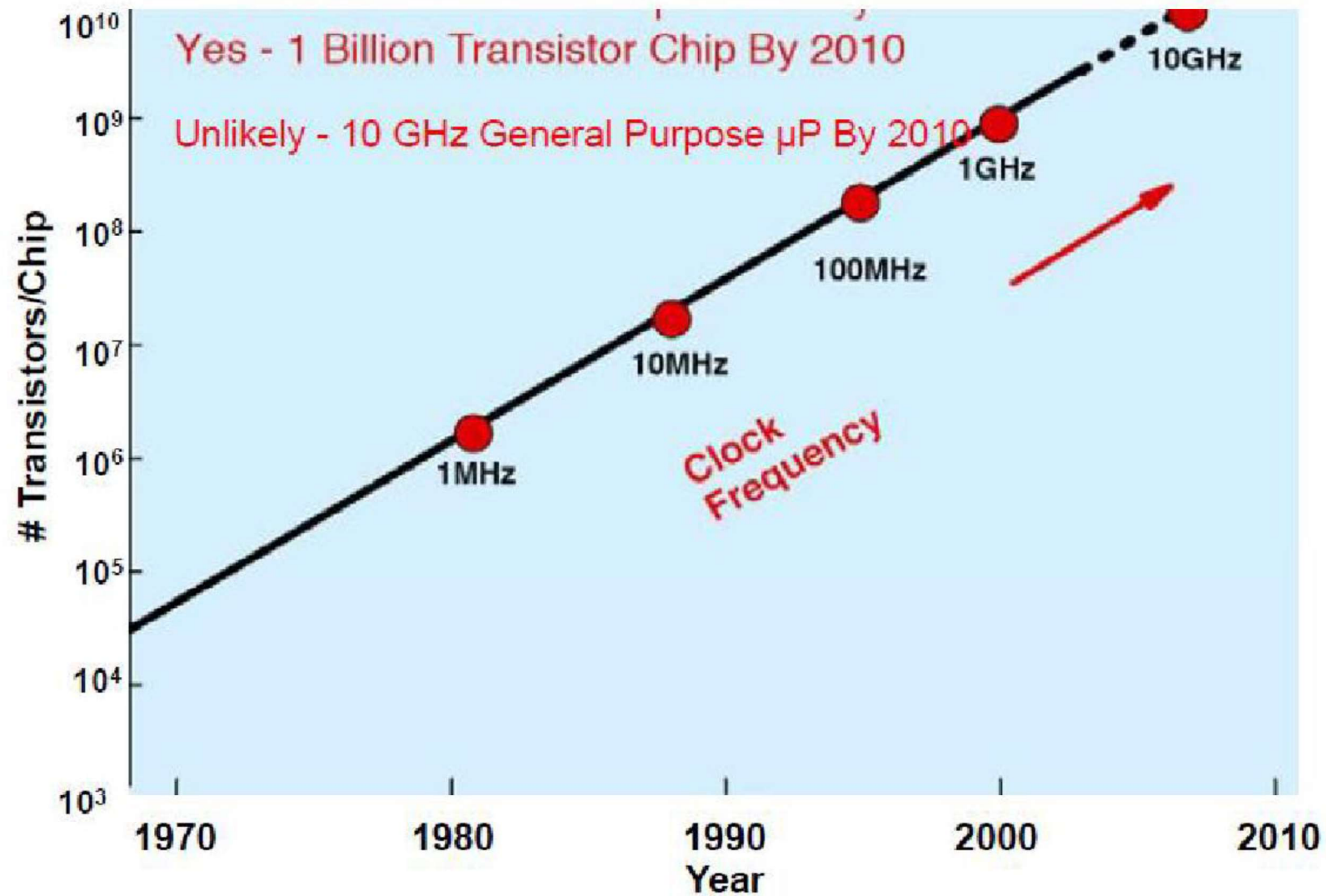
Moore's law (what didn't work)



Cost argument of Moore's law is dead...



Moore's law (what didn't work)



Is Moore's Law Dead?

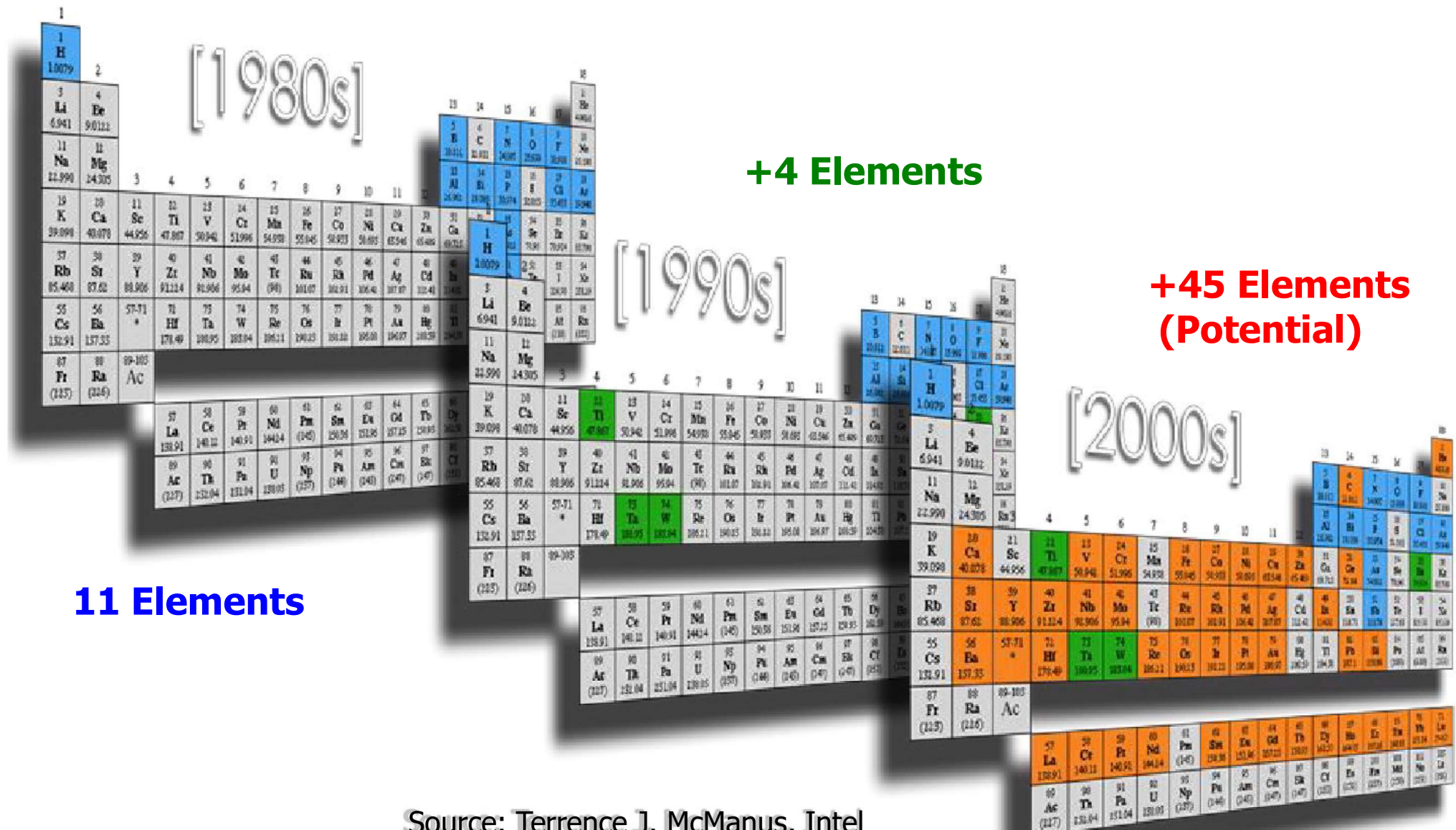
- ❑ Still valid but with a slower pace
- ❑ Going to be obsolete (estimated in 2020s)
- ❑ Threads to Moore's Law
 - Size approaching atomic scale (Heisenberg's uncertainty principle)
 - Growing cost (design cost for a new 10 nm chip is around \$170 million, almost \$300 million for a 7 nm chip and over \$500 million for a 5 nm chip)

What will be beyond Moore's Era?

- ❑ Quantum Computing (based on qubits or quantum bits)
- ❑ Specialized Architecture tuned to particular algorithms (e.g., GPU for AI training)
- ❑ Spintronics computing
- ❑ Optical computing (light is manipulated using photodetectors, phase modulators, waveguides etc.)
- ❑ Neuromorphic computing

It's a material world !

Material Complexity Increasing Exponentially



Source: Terrence J. McManus, Intel

Semiconductor Devices and Materials

- Chips – Logic, Memory, Cellphones, ASICs, Sensors, Health Care, Displays, RFID tags, Entertainment
 - Silicon (in-production)
 - Germanium, III-V, Carbon (in-research)
- Wireless Circuits – Agilent (Cellphones, Routers)
 - III-V (in production)
- LEDs
 - III-V (in-production)
- Solar Cells – Si, III-V, CIGS (in-production)

Electronics vs. Photonics

Electronics

- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si/SiO₂/PolySi
- Si electronics is a **processing** based technology

Photonics

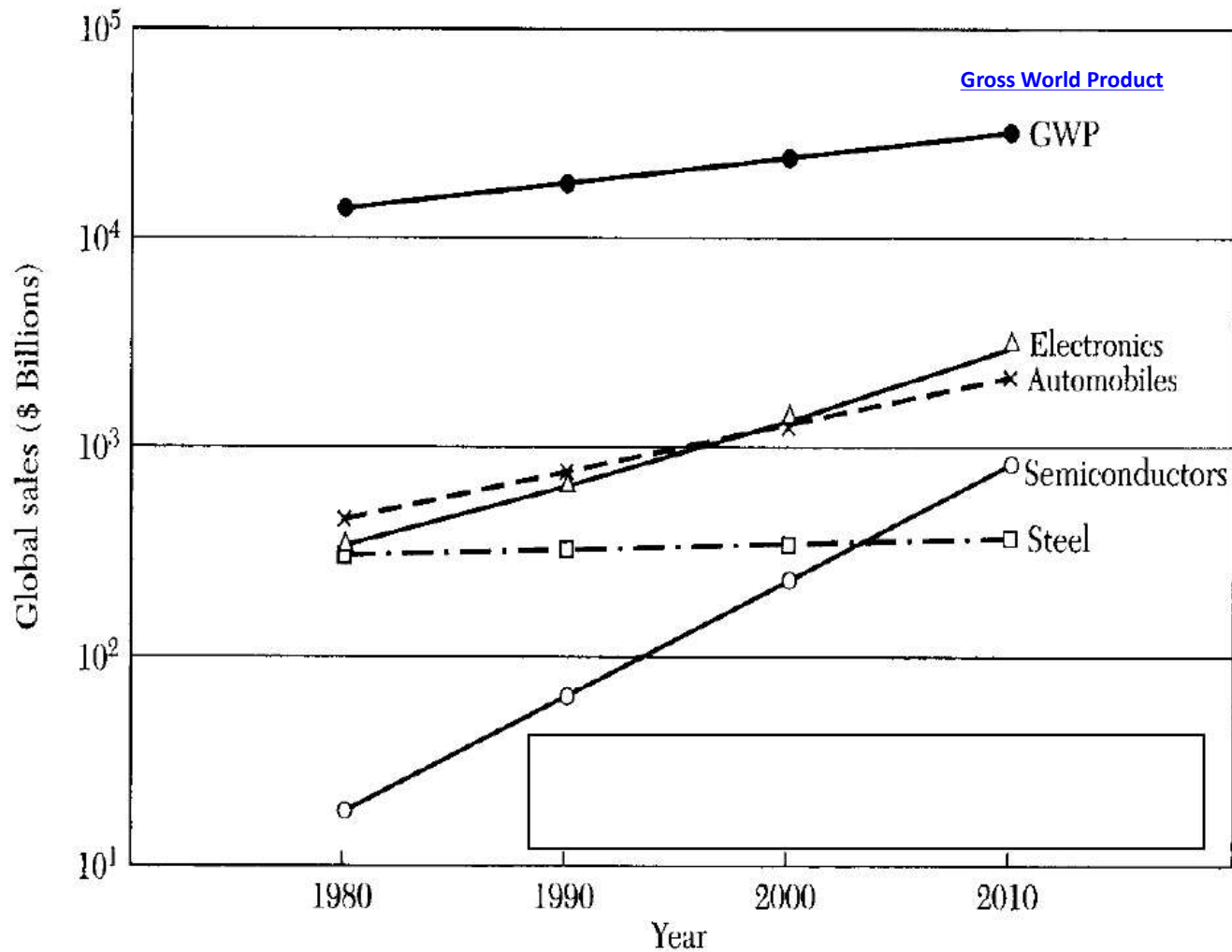
- GaAs, GaP, GaSb, InP, InAs, InSb, AlAs, GaN, AlN
- GaAs_xP_{1-x}, Al_xGa_{1-x}As, In_xGa_{1-x}P, In_xGa_{1-x}As, Al_xGa_{1-x}Sb, Al_xGa_{1-x}N, Ga_xIn_{1-x}N--**Ternary Alloys**
- In_xGa_{1-x}As_yP_{1-y}, Al_xIn_yGa_{1-x-y}As, In_xGa_{1-x}As_ySb_{1-y}, In_xGa_{1-x}N_yAs_{1-y}--**Quaternary Alloys**
- In_xGa_{1-x}N_yAs_{1-y-z}Sb_z--**Quinary Alloy**
- Heterojunctions are required for carrier confinement
- Photonics is a **materials** based technology

LED Color and Materials

Color	Wavelength (nm)	Voltage (V)	Semiconductor material
Infrared	$\lambda > 760$	$\Delta V < 1.9$	Gallium arsenide (GaAs) Aluminium gallium arsenide (AlGaAs)
Red	$610 < \lambda < 760$	$1.63 < \Delta V < 2.03$	Aluminium gallium arsenide (AlGaAs) Gallium arsenide phosphide (GaAsP) Aluminium gallium indium phosphide (AlGaInP) Gallium(III) phosphide (GaP)
Orange	$590 < \lambda < 610$	$2.03 < \Delta V < 2.10$	Gallium arsenide phosphide (GaAsP) Aluminium gallium indium phosphide (AlGaInP) Gallium(III) phosphide (GaP)
Yellow	$570 < \lambda < 590$	$2.10 < \Delta V < 2.18$	Gallium arsenide phosphide (GaAsP) Aluminium gallium indium phosphide (AlGaInP) Gallium(III) phosphide (GaP)
Green	$500 < \lambda < 570$	$1.9^{[47]} < \Delta V < 4.0$	Indium gallium nitride (InGaN) / Gallium(III) nitride (GaN) Gallium(III) phosphide (GaP) Aluminium gallium indium phosphide (AlGaInP) Aluminium gallium phosphide (AlGaP)
Blue	$450 < \lambda < 500$	$2.48 < \Delta V < 3.7$	Zinc selenide (ZnSe) Indium gallium nitride (InGaN) Silicon carbide (SiC) as substrate Silicon (Si) as substrate — (under development)
Violet	$400 < \lambda < 450$	$2.76 < \Delta V < 4.0$	Indium gallium nitride (InGaN)
Purple	multiple types	$2.48 < \Delta V < 3.7$	Dual blue/red LEDs, blue with red phosphor, or white with purple plastic
Ultraviolet	$\lambda < 400$	$3.1 < \Delta V < 4.4$	Diamond (235 nm) ^[48] Boron nitride (215 nm) ^{[49][50]} Aluminium nitride (AlN) (210 nm) ^[51] Aluminium gallium nitride (AlGaN) Aluminium gallium indium nitride (AlGaInN) — (down to 210 nm) ^[52]
White	Broad spectrum	$\Delta V = 3.5$	Blue/UV diode with yellow phosphor

Next growth drivers

Semiconductor Industry



2011	Revenue (B \$)
Intel	50
Samsung	30
Qualcomm	11
nVidia	4

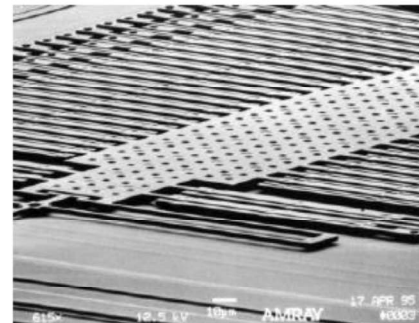
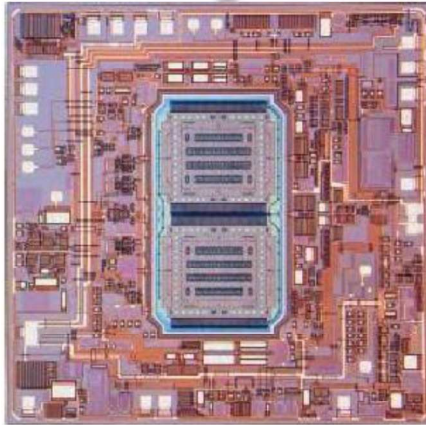
No 2 export of the USA

What's enabling these ?

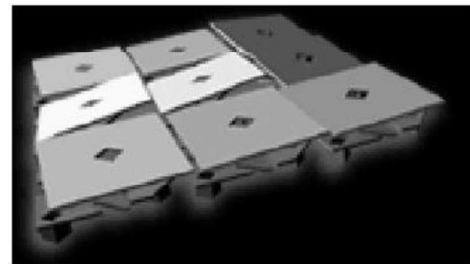


MEMS ?

~ MEMS: a \$7B market in 2009, growing until 2007 at 15%/year

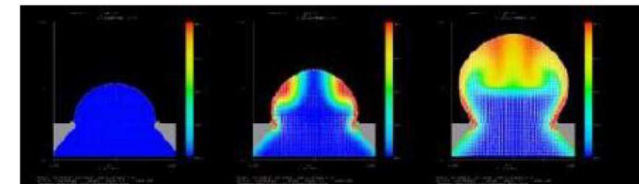
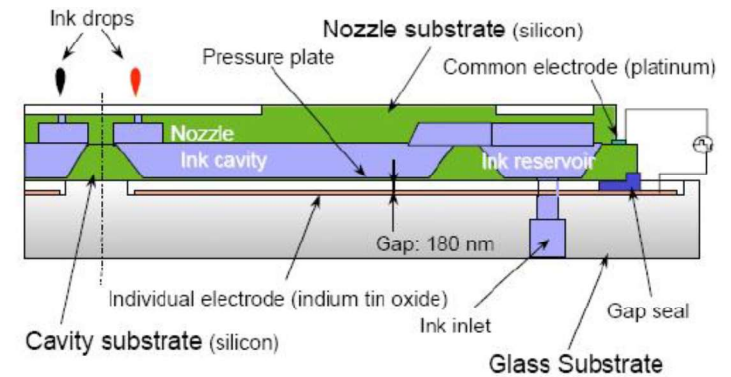


Inertial Sensors
(Analog Devices)



Micromirror Projectors
(Texas Instruments DLP™)

Cross-Sectional View of SEAJet^(R) Printhead



Ink Jet Printheads
Epson (top), Kodak (bottom)

Photonic integrated circuit (PIC)



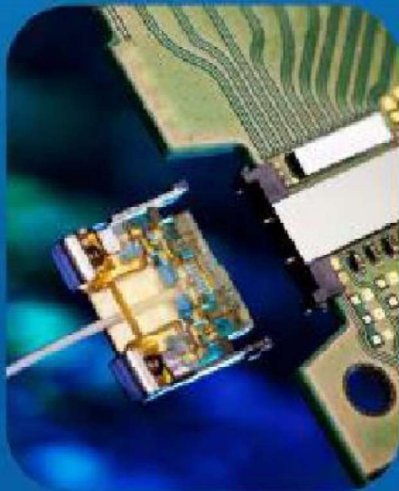
Why Photonics?

Copper wires reaching physical limits

- ~10 Gbps or higher becoming challenging
- Distance/speed tradeoff shortens lengths

Alternative: Transmit data over optical fiber

- Much further reach at any given speed
- Multiple signals can travel on one fiber
- Thin & light = easy cable management



Challenge: optical signaling technology is expensive



PhotoVoltaics (PV)



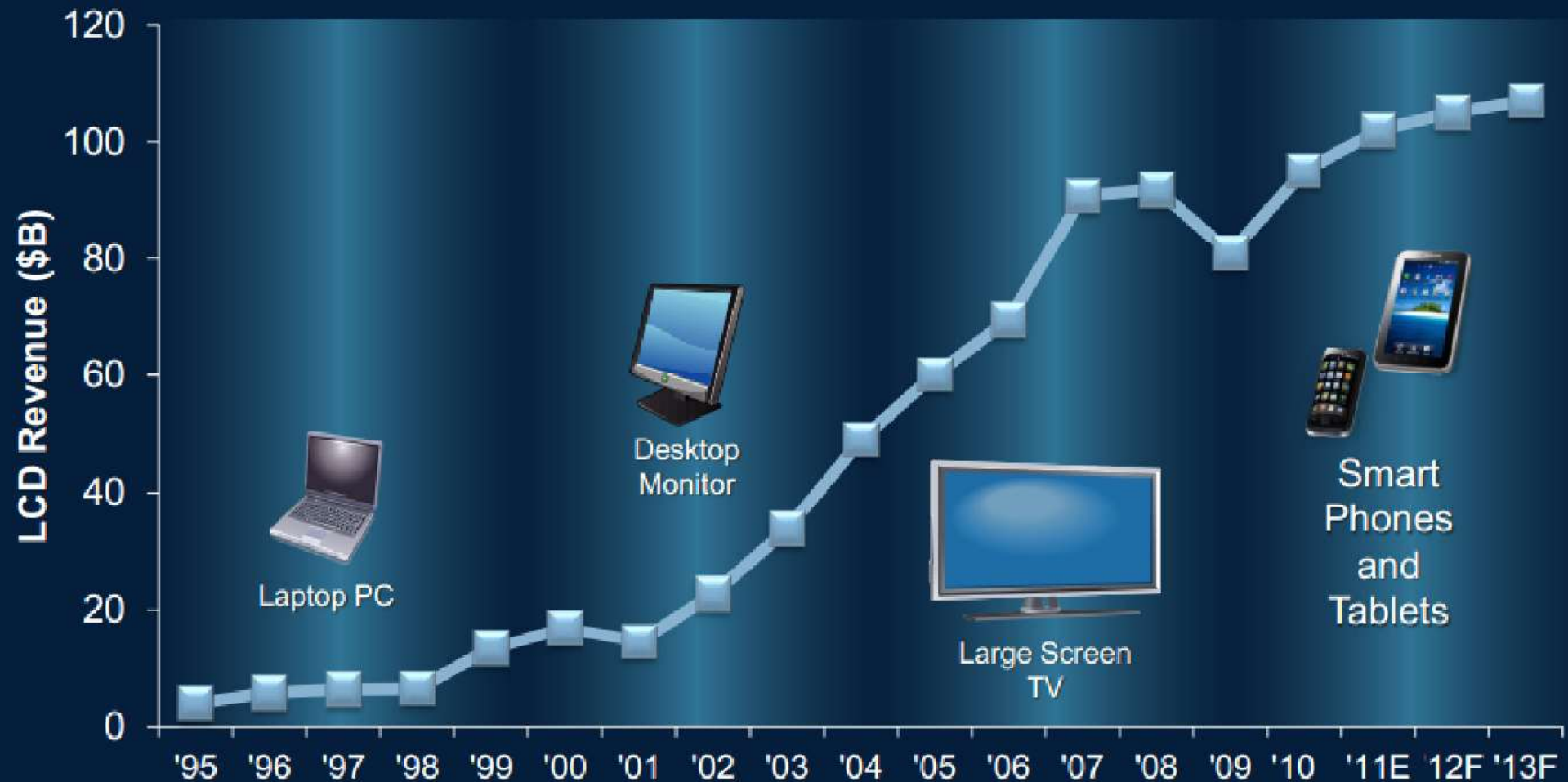
Si, III-V, CIGS

Issues: \$\$, Weather conditions, Batteries, DC to AC losses.

Massive investments: 1970s, 2000s. Likely to play a niche role, in a huge and ever-growing part of the world economy.

Display

Mobility Creating “Fourth Wave” in Displays



- Better image quality needed for new mobile content
- Multi-touch interface needed for flexibility w/o mouse and keyboard